

## VESPER-SEC-01 — Week 6 Progress Report

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### What Was Attempted

Built the Week 6 charter deliverable: a minimum working benchmark artifact for four cryptographic primitives (Ascon v1.2, ChaCha20-Poly1305, ML-KEM/Kyber-512, SPHINCS+-Haraka-128f), compiled for ARM Cortex-M4 and executed under Renode, using the DWT cycle-counter methodology defined in the Week 5 analytical model.

### What Worked

| Item                               | Result                                                                               |
|------------------------------------|--------------------------------------------------------------------------------------|
| Ascon v1.2                         | 9,606 cycles (16 B) — 600.375 cycles/byte; stack 264 B; ELF 23,028 B                 |
| ChaCha20-Poly1305                  | 40,597 cycles (114 B) — $\approx$ 356.114 cycles/byte; stack 544 B; ELF 5,252 B      |
| Kyber-512                          | KeyGen 1,157,443 / Enc 1,300,194 / Dec 1,538,934 cycles; stack 9,748 B; ELF 14,280 B |
| SPHINCS+-Haraka-128f               | KeyGen 153,849,855 / Sign 3,738,100,285 / Verify 242,116,523 cycles; ELF 28,888 B    |
| Analytical limit (carried forward) | 1,000-NOP test: 1,003 measured vs. 1,000 expected (0.3% discrepancy)                 |

### Test Suite Verification

A project-specific Week 6 test suite was executed after completion of the benchmark artifacts. The suite contains 22 checks covering the analytical limiting case and logical/result checks for Ascon v1.2, ChaCha20-Poly1305, ML-KEM/Kyber-512, and SPHINCS+-Haraka-128f.

**Test result: 22/22 Week 6 project test-suite checks passing.**

The analytical limiting-case check measured 1,003 cycles for the 1,000-instruction NOP test, corresponding to a 0.3% discrepancy from the expected value.

*Test scope: these tests verify the Week 6 analytical/logic conditions and recorded benchmark outputs. They do not constitute published-result reproduction or cryptographic cycle-accuracy validation. The SPHINCS+ peak-stack measurement was excluded because it was identified as invalid.*

## What Did Not Work / Open Items

SPHINCS+ peak-stack measurement was invalidated — the stack-pattern region had already been overwritten before the measurement scan, causing the raw 57,856 B figure to represent the full scanned region rather than a valid peak-stack measurement. The figure is therefore excluded from stack comparison.

The Heinz [8] reference was successfully built after correcting missing `.type/.thumb_func` metadata in four Keccak assembly functions. During review, we confirmed that Heinz implements first-order masked Kyber768, not Kyber-512. Since this differs from the project's unmasked Kyber-512 parameter set, no direct cycle-count comparison was made. Published-result validation is outside the Week 6 scope and is planned for the later validation stage.

### Changes to the Plan

No change was made to the completed Week 6 benchmark results. During review, a documentation correction was identified: the Week 4 proposal described Heinz [8] as Kyber-512, whereas the investigated Heinz implementation is first-order masked Kyber768. No Week 6 benchmark result was changed on the basis of this reference.

### Next Week's Specific Objective (Week 7)

Per the charter, Week 7 is the extension stage. The next objective is to extend the Week 6 working artifact with the capability specified in the Week 7 charter, without changing the recorded Week 6 measurements.